MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 4DP7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Square face, two gun, electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	Eb3	ed	Rg	Zd	Ehk	Eb3/Eb2
Unit:	V	Vdc	Vdc	Vdc (Note 1)	Vdc	v	Meg	Meg (Note 2)	Vdc	Ratio (Note 3)
Maximum:	6.9	0	1,500	3,000	6,000	750	1.5	1.0	±180	2.0
Minimum:	5.7	-200	---	1,000	2,000	---	---	---	---	2.0
TEST CONDITIONS: (Note 4)	6.3	Adjust	Focus	2,000	4,000	---	---	---	---	2.0

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
5111	Vibration	-		---	---	1.0	mm
(D) 5216	Cathode illumination	7		---	---	0.154	mftL
(D) 5101	Neck and bulb alinement (electrostatic types)	-		---	---	3.2	Inch
(D) 5248	Deflection-factor uniformity	-		---	---	2.5	%
(D) 1331	Direct-interelectrode capacitance	11	g1 to all k to all D1 to D2 D3 to D4 D1 to all D2 to all D3 to all D4 to all	Cg1 Ck C1D2 C3D4 CD1 CD2 CD3 CD4	--- --- --- --- --- --- --- ---	6.0 5.0 2.3 2.3 5.3 5.3 6.5 6.5	pF pF pF pF pF pF pF pF
(D) 1141	Pressure (implosion)	-		---	---	---	---
(D) 5250	Interaction factor	-		---	---	$5X10^{-5}$	in./V
(D) 1216	Base material insulating quality	-		---	---	---	---
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	14	Ib3 = 25 µAdc	---	---	---	---
5106	Screen and faceplate blemishes	-		---	---	---	---
5223	Modulation	-	Ib3 = 25 µAdc	ΔEc1	---	30	Vdc
5231	Spot position (electro-static deflection)	9		---	---	13	mm
5231	Spot displacement (leakage)	-		Displ	---	5	mm
5241	Grid cutoff voltage	-		Ec1	-52	-87	Vdc
---	Tracking error	12, 13		---	---	0.043	Inch
---	Minimum useful scan	10		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-		Ib1	-15	+5	µAdc
5201	Electrode current (cathode)	-	Ib3 = 25 µAdc	Ik	---	200	µAdc
5101	Base alinement (electrostatic types)	-	+1D2; pin No. 4	---	---	---	---
5101	Side terminal alinement (electrostatic types)	-	+1D2	---	---	---	---
5101	Side terminal and base alinement (electrostatic types)	-	Pin No. 4	---	---	---	---

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METHOD		REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
		Quality conformance inspection, part 2 -Continued						
	5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
(D)	---	Trace and bulb alinement (electrostatic types)	5		---	---	1	Degree
(D)	5101	Angle between traces	-		---	89.5	90.5	Degree
	---	Trace alinement	6		---	---	1	Degree
(D)	---	Trace curvature radius			Rad	40	---	Inches
	5216	Stray light emission	-	Eb2 = 3,000 Vdc; Eb3 = 6,000 Vdc	---	---	---	---
	5221	Screens	8		---	---	---	---
	5226	Line width "A" (electrostatic deflection)	-	Ib3 = 25 µAdc	Width	---	0.5	mm
	5226	Line width "B" (electrostatic deflection)	-	Ib3 = 25 µAdc	Width	---	0.6	mm
	5246	Focusing voltage at cutoff	-		Eb1	---	550	Vdc
	5246	Focusing voltage at modulation condition	-		Eb1	350	---	Vdc
	5248	Deflection factor (1D2)	-		DF	115	140	Vdc/in.
	5248	Deflection factor (3D4)	-		DF	90	115	Vdc/in.
	5251	Heater-cathode leakage current	-		---	---	15	µAdc
	5251	Grid No. 1 leakage current	-		---	---	3	µAdc
	5251	Anode No. 1 leakage current	-		---	---	3	µAdc
	5251	Anode No. 2 leakage current	-		---	---	5	µAdc
	1101	Secureness of base, cap, or insert	-		---	---	---	---
(D)	1111	Base pin solder depth	-		---	---	---	---
(D)	1105	Permanence of marking	-		---	---	---	---
		Quality conformance inspection, part 3						
	---	Life test	-	Group C; t = 500 hours (min); Ib3 = 15 µAdc; Eb2 = 3,000 V; Eb3 = 6,000 V; both guns operating	---	---	---	---
	---	Life-test end points:						
	5223	Modulation	-	Ib3 = 20 µAdc	ΔEc1	---	30	Vdc
	5226	Line width "A"	-		Width	---	0.5	mm
	5226	Line width "B"	-		Width	---	0.6	mm

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NOTES:

1. Accelerator power input (average) should be limited to 6 watts.
2. It is recommended that the deflection electrode circuit resistance be approximately equal. Higher resistance values up to 5.0 megohms may be used for low-beam current operation.
3. This tube is designed for optimum performance when operating at an Eb3/Eb2 ratio of 2.0. Operation at other ratios of Eb3/Eb2 may result in changes in deflection uniformity, pattern distortion, and tracking accuracy.
4. All tests except capacitance, vibration, and interaction factor shall be made on each unit separately. Each deflection plate shall be connected to A2 through a resistance of 1 megohm.
5. Alinement between any trace and the parallel edge of the faceplate shall not exceed the indicated value.
6. The angle between corresponding traces of each gun shall not exceed the indicated value.
7. The measurement shall be made with both filaments operating simultaneously.
8. The raster size shall be 4.1 x 4.1 cm. The screen current shall be adjusted to 20 µAdc. Add 48 cB to the reading to correct for the smaller raster area.
9. With focused, undeflected spot and the tube shielded against external influences the spots will fall within a 13 mm square, the center of which coincides with the geometric center of the tube face and the sides of which are parallel to the traces produced by the various deflecting electrodes.
10. The useful scan will be a 2-7/8 (73.03 mm)-inch square having rounded corners. The maximum radius of curvature of the rounded corners of the square screen area shall be 0.6 (15.24 mm) inch.
11. All other electrodes contained within the envelope are to be tied together.
12. The measurement of tracking error shall be made as follows:
    The focused spots of both guns shall be moved by means of a common dc voltage along the 1D2 axis until the spot of gun A is at +23/32(+18.26 mm) inch from the geometrical center of the tracking area. Then by means of a small correction voltage the spot of gun B shall be moved to the same distance. The spots then shall be moved by means of the common dc voltage in the opposite direction from the geometrical center until the spot of the less sensitive gun reaches -23/32 (-18.26 mm) inch. The difference in deflection factor of the more sensitive gun is then compensated for by moving its spot back to -23/32 (-18.26 mm) inch by means of a voltage divider. This entire procedure is to be duplicated along the 3D4 axis with a distance from the geometrical center of ±9/16 (±14.29 mm) inch. Then by means of the two common dc voltages the spots shall be tracked over the entire tracking area, a 2-7/8 (73.03 mm) inch (1D2) by 2-1/4 (57.15 mm) inches (3D4) rectangle, and the greatest distance between the spots shall not exceed the limit specified.
13. The maximum distance along the vertical or horizontal axis shall not exceed the value specified.
14. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

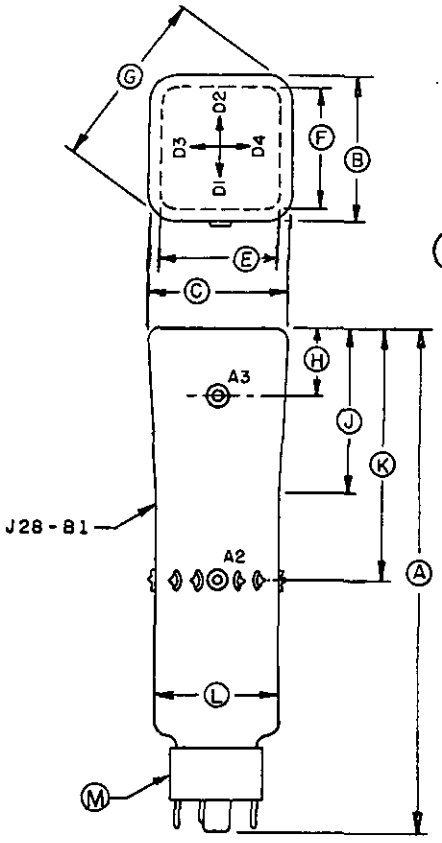
Preparing activity:
Air Force - 85

Review activities:
Air Force - 17, 99
DSA - ES

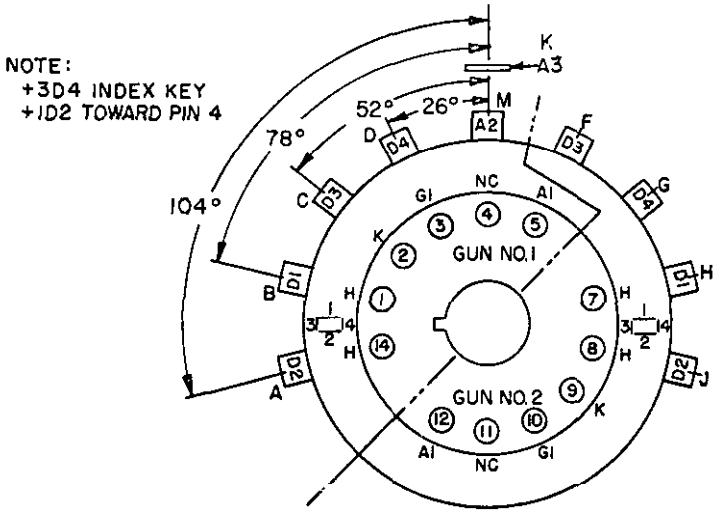
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User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 11

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
D	Qualification inspection	
M	Base: B12-37	
	Quality conformance inspection, part 2	
A	12.000 (304.80)	12.500 (317.50)
B	3.437 (87.30)	3.563 (90.50)
C	3.437 (87.30)	3.563 (90.50)
E	2.875 (73.02)	
F	2.875 (73.02)	
G	4.156 (105.56)	4.344 (110.34)
H	1.500 (38.10)	1.750 (44.45)
J	4.125 (104.78)	4.875 (123.83)
K	6.000 (152.40)	6.375 (161.93)
L	2.969 (75.41)	3.031 (76.99)



BOTTOM VIEW OF
BASE CONNECTIONS

FIGURE 1. Outline drawing of electron tube type 4DP7.

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